MDC-FusFormer: Multiscale Deep Cross-Fusion Transformer Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*— The spatial resolution of hyperspectral images (HSIs) is usually limited due to internal imaging mechanisms. To obtain imagery with high spectral and high spatial resolutions, which is essential for subsequent HSI processing tasks, a cost-effective approach is to fuse HSI with multispectral images (MSIs). One highly effective fusion method is the convolutional neural network (CNN). However, CNNs have limitations in capturing global information and complex features. Recently, visual transformers (ViTs) have garnered interest for their ability to process non-local information. Despite this, existing HSI-MSI fusion methods suffer from insufficient spatial–spectral feature interaction, resulting in suboptimal fusion quality. To address these challenges, we propose a multiscale deep cross-fusion transformer (MDC-FusFormer) network for HSI and MSI fusion. This network effectively performs the interactive fusion of spatial–spectral features, thereby enhancing the quality of the fused images. MDC-FusFormer employs a three-branch network architecture consisting of two independent progressive feature mining modules (PFMMs), a multiscale deep cross-fusion attention module, and a spatial–spectral feature fusion module. Initially, shallow features at different scales of MSI and HSI are recursively extracted through successive up- and down-sampling using CNNs. These features then interact with the deep cross-modal information at corresponding scales through the attention block. Finally, a multidimensional refinement convolution block (MRCB) is applied to refine the feature information, which is then combined with cascaded up-sampling to reconstruct the high-resolution fused image step by step. Experimental results on five datasets indicate that, compared to nine other methods, MDC-FusFormer delivers superior performance.**

***Index Terms*— Convolutional neural network (CNN), cross-attention mechanism, hyperspectral and multispectral image fusion, transformer.**

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I. Introduction

DUE to the advancement in hyperspectral imaging technology, hyperspectral images (HSIs) with high spectral resolution can be acquired, significantly facilitating the identification and acquisition of geomorphological information. According to the characteristics of HSI [1], [2], it is widely used in the fields of ecosystem monitoring, mineral survey, plant survey, military monitoring, and early disaster warning. Nevertheless, the processing of information is largely dependent on the quality of the images. Unfortunately, due to limitations in imaging cameras, HSIs often suffer from low spatial resolution, which poses a challenge to the widespread application of hyperspectral imaging technology. Consequently, acquiring high-resolution images has become a significant focus in the field of hyperspectral image processing. Currently, the fusion of HSI and multispectral image (MSI) stands out as one of the more promising methods to enhance image resolution in both the spectral and spatial domains.

Existing HSI and MSI fusion methods can be broadly categorized into traditional methods and deep learning-based methods. Traditional methods can be categorized into methods based on matrix factorization (MF) [3], [4], [5], [6], [7], [8] and tensor factorization (TF) [9], [10], [11]. MF-based methods mainly fuse images by unfolding and decomposing the target high-resolution HSI (HR-HSI) into matrices along the spectral dimension. The method assumes that the matrix can be decomposed into the product of a spectral basis matrix and its coefficients. At the same time, both HR-MSI and the low-resolution hyperspectral image (LR-HSI) are obtained from the subsampling of HR-HSI. Usually, the spectral basis represents the spectral information of the HR-HSI. Based on the optimization formulas of spectral basis and coefficients, matrix decomposition-based approaches can

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generally be classified into three categories. The first class of methods aims to obtain the spectral information from the LR-HSI only, that is, the spectral basis is estimated only from the observed HSI, and the coefficients are obtained from the MSI. For example, Kawakami et al. [12] introduced sparse MF (SMF) for HSI-MSI fusion. The spectral dictionary was estimated using a sparse dictionary learning method and the coefficients were estimated using a sparse coding algorithm on MSI to reconstruct the spectrum at each location. Huang et al. [13] proposed a similar method, named the spatial and spectral fusion model (SASFM), which utilized K-singular value decomposition (K-SVD) to learn the spectral dictionary. Subsequently, the method employed sparse coding to reconstruct the spectrum for each pixel, thereby producing the fused image. This method effectively maintained precise spectral estimation while markedly improving the spatial resolution. Akhtar et al. [14] proposed a generalized Bayesian sparse coding framework for the construction of HR-HSI by combining the computed codings with estimated spectra from a Bayesian model. The second class of methods is based on the assumption that spatial information is also contained in the LR-HSI. Based on this assumption, the spectral basis is calculated from the LR-HSI and then the coefficients are calculated from both the HSI and MSI. Wei et al. [15] proposed a patch-based sparse a priori method to maintain self-similarity and solve the fusion problem by alternately optimizing the target image and the coding coefficients. After that, Wei et al. [16] proposed a fast multiband image fusion algorithm for HSI and MSI fusion, also known as a fast fusion algorithm based on Sylvester's equations (FUSE), which greatly reduced the computation time. The third class of methods uses a coupling matrix decomposition-based approach to update the spectral basis and the coefficients alternately. For instance, Yokoya et al. [4] introduced coupled non-negative MF (CNMF) into HSI-MSI fusion. The fused images obtained by the unmixed CNMF-based approach had smaller spectral aberrations while improving the spatial resolution of the images in all hyperspectral bands. To make full use of the acquired HSI and MSI, Shen et al. [17] proposed a twice-optimized network with matrix decomposition to perform the fusion task and obtain better results. Lin and Guo [18] proposed a highly interpretable deep HSI-MSI fusion method based on probabilistic MF (PMF) in a Bayesian framework, which greatly improved the generalization ability and image quality. However, they are prone to information loss in the process of fusion and often need to solve complex optimization problems iteratively, which is computationally intensive.

TF-based methods treat an image as a 3-D tensor to ensure its spatial and spectral properties. Tucker decomposition [19] is a widely used method in TF. It involves decomposing a high-dimensional tensor into a factor matrix of each dimension and a core tensor, which can separate the information of each dimension in each tensor and simultaneously establish the correlations through the core tensor. Li et al. [20] proposed an HSI-MSI fusion method based on coupled sparse TF (CSTF), which defined the fusion problem as the estimation of the core tensor and the dictionaries of the three modes but did not take into account the non-local spatial similarity. Prévost et al. [21] used the truncated SVD to obtain the dictionaries of the three modes and utilized the generalized Sylvester equation to estimate the core tensor. This significantly reduced the computational cost. In [22], a non-local sparse tensor decomposition, also known as non-local sparse tensor factorization for semi-blind fusion (NLSFT-SMBF), was proposed for HSI-MSI fusion. Guo et al. [23] proposed a coupled non-negative block-term tensor decomposition model, which coupled the block matrix and the endmember matrix together to improve the fusion performance. This approach also introduced non-local spatial similarity into the sparse tensor decomposition and made full use of the spatial–spectral information of HSI and MSI. TF-based methods excel in preserving the structural integrity and information content of the data when compared to MF-based approaches; however, they are more prone to being influenced by heavy noise and typically have higher computational complexities.

The majority of these conventional methods depends on a priori modeling of the image, is parameter-choice sensitive, and heavily relies on precise estimations of the point spread function (PSF) and the spectral response function (SRF). As deep learning has progressed, all parameters can now be learned directly from the training data by deep networks, without the need for any a priori assumptions about the image. Additionally, convolutional neural networks (CNNs) [24], [25], [26], [27], [28] have demonstrated exceptional prowess in feature extraction. For instance, Dong et al. [29] first proposed a deep convolutional network, namely the super-resolution CNN (SRCNN), for single image super-resolution, and achieved good performance. Pan-sharpening is a special case of HSI-MSI fusion. Initially, the spatial–spectral fusion of LR-MSI and panchromatic (PAN) images [30], [31], [32], [33] to improve the spectral and spatial resolution of the images was known as panchromatic sharpening. With the development of hyperspectral imaging technology, the panchromatic sharpening technique was gradually extended to the field of HSI and MSI fusion. Chen et al. [34] proposed a panchromatic sharpening method that divided the entire spectrum covered by the hyperspectral band into several regions. This method interpolated the missing band images into MSI and finally fused the HSI and MSI of each region with less spectral distortion. Grohnfeldt et al. [35] proposed a pan-sharpening algorithm based on spectral grouping to interpret HSI-MSI fusion as a weighted panchromatic sharpening problem using the relative spectral responses of the two instruments. Yuan et al. [36] proposed a multiscale and multidepth CNN (MSDCNN) for pan-sharpening and achieved good performance. However, the direct application of conventional full-color sharpening techniques to the fusion of HSI and MSI does not yield optimal results. Therefore, Dian et al. [37] proposed a deep HSI pan-sharpening method, called deep HSI sharpening (DHSIS). This approach leveraged residual learning within deep CNNs to directly learn image priors, thus enhancing fusion accuracy and reducing processing time. Cai and Huang [38] introduced a super-resolution module and progressive learning in a network that continuously captured spatial details at different scales and injected them

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into the MSI. Zhang et al. [39] proposed an interpretable spatial–spectral reconstruction network (SSR-NET) with spatial and spectral loss functions, achieving competitive fusion results. Addressing the challenge of non-overlapping spectral ranges in the HSI-MSI fusion, Sun et al. [40] proposed a band divide-and-conquer framework (BDCF) that thoroughly considered spectral fidelity, spatial enhancement, and computational efficiency. Subsequently, to effectively fuse spatiotemporal HSI and high spatiotemporal multispectral images (MSIs), Sun et al. [41] proposed an innovative unsupervised 3-D tensor subspace decomposition network. The network used a 3-D HSI/MSI tensor subspace extraction network to predict missing low spatial resolution hyperspectral tensor features at other times, reconstructing remote sensing images with enhanced temporal, spatial, and spectral resolutions. Wang et al. [42] tackled the issue of up-sampling's inability to fully restore continuous spatial and spectral information by introducing a spatial–spectral implicit neural representation network for HSI and MSI fusion. However, most deep learning approaches rely on CNNs to extract local feature information from the two images, often overlooking the global context. This oversight results in a fusion where spatial and spectral features lack interaction and fail to achieve a strong coupling. Additionally, the typical CNN architecture encompasses numerous network layers, leading to extended training durations.

Since 2020, a new visual transformer (ViT) [43] has made a big splash in the field of computer vision (CV), particularly in areas such as image classification [44], [45], semantic segmentation, image generation, and target detection. Due to its self-attention mechanism for acquiring long-range information, which solves the problem that CNNs are not good at extracting global information, some scholars have proposed applying the transformer model to fusion tasks. Bose et al. [46] were the first to employ the transformer architecture in the domain of HSI fusion, successfully integrating HSI and LiDAR data to facilitate effective cross-modal information exchange. Then, Hu et al. [47] proposed a transformer-based fusion method (FusFormer), which globally explored the connection between the HSI and MSI feature internals and achieved excellent fusion performance. You et al. [48] proposed a U-shaped transformer structure for HSI-MSI fusion (HMF-Former). Through multihead attention, the spatial correlation of MSI and the spectral self-similarity of HSI were extracted separately. Jia et al. [49] proposed a multiscale spatial–spectral transformer network (MSST-Net) to address its difficulty in training on small datasets of the problem. The self-attentive mechanism performs well in modeling long-range dependencies since it can effectively extract global context information. Deng et al. [50] proposed a pyramid shuffle-and-reshuffle transformer (PSTR) network for HSI-MSI fusion. This approach solved the computational complexity problem by calculating the self-attention of a smaller set of fixed-size windows. Sun et al. [51] proposed a spatial–spectral pyramid network (DSPNET) for the fusion of two images which focused on the extraction of spectral information and local spectral information through the spectral pyramid structure. Ma et al. [52] proposed a dual-cross transformer-based fusion method (DCTransformer) that integrated spatial and spectral fusion. Nowadays, most transformer-based HSI-MSI fusion methods perform feature extraction for individual modality and then perform feature concatenation for fusion. Or, two images are preliminarily concatenated first, and then the concatenated images or blocks are input into the transformer to capture non-local dependencies. This makes it difficult for the transformer to capture effective non-local complementary information, which leads to insufficient cross-modal information interaction between spatial–spectral features and lower fusion accuracy. Moreover, the full utilization of spatial information is not achieved, which is detrimental to the fusion task.

Therefore, this article proposes a new multiscale deep cross-fusion transformer (MDC-FusFormer) network for HSI and MSI fusion. The MDC-FusFormer consists of three main components: two independent progressive feature mining modules (PFMMs) for HSI and MSI, a multihead deep cross-fusion attention (MDCFA) module and a spatial–spectral reconstruction module (SSFM), respectively. The network employs a three-branch network structure. Firstly, the PFMM is used to extract the features at different scales of HSI and MSI as well as MSI after successive up- and down-sampling. Then, the feature maps at the same scales are input into MDCFA where correlations are calculated through pairwise interactions, facilitating cross-spatial–spectral information exchange. Lastly, a multidimensional refinement convolution block (MRCB) in SSFM is designed to enhance spatial and spectral details in three dimensions (i.e., horizontal, vertical, and channel), in conjunction with cascaded up-sampling to reconstruct HR-HSI. The main contributions of this article are summarized as follows.

1) A novel network structure (MDC-FusFormer) is proposed for the HSI-MSI fusion task. The HR-HSI is reconstructed sequentially through a PFMM, an MDCFA module, and an SSFM.
2) A multibranch cross-feature attention fusion module is proposed for deep cross-fusion of spatial and spectral features in MDC-FusFormer. The module makes full use of spatial information by adding a processed MSI to capture global cross-spatial and spectral dependencies and long-range details for the effective interaction of spatial and spectral features.
3) A multidimensional refinement convolution module is proposed to reconstruct high-resolution images, which captures spatial details of different dimensions from the horizontal, vertical, and channel dimensions through specific transformations, thereby significantly improving the fusion performance.
4) Compared with nine state-of-the-art methods, the proposed MDC-FusFormer achieves competitive results on five classical datasets. Experimental results demonstrate the superiority of the proposed MDC-FusFormer in HSI-MSI fusion.

The remainder of the article is organized as follows. In Section II, the proposed MDC-FusFormer model is elaborated. Section III addresses the extensive experimental results

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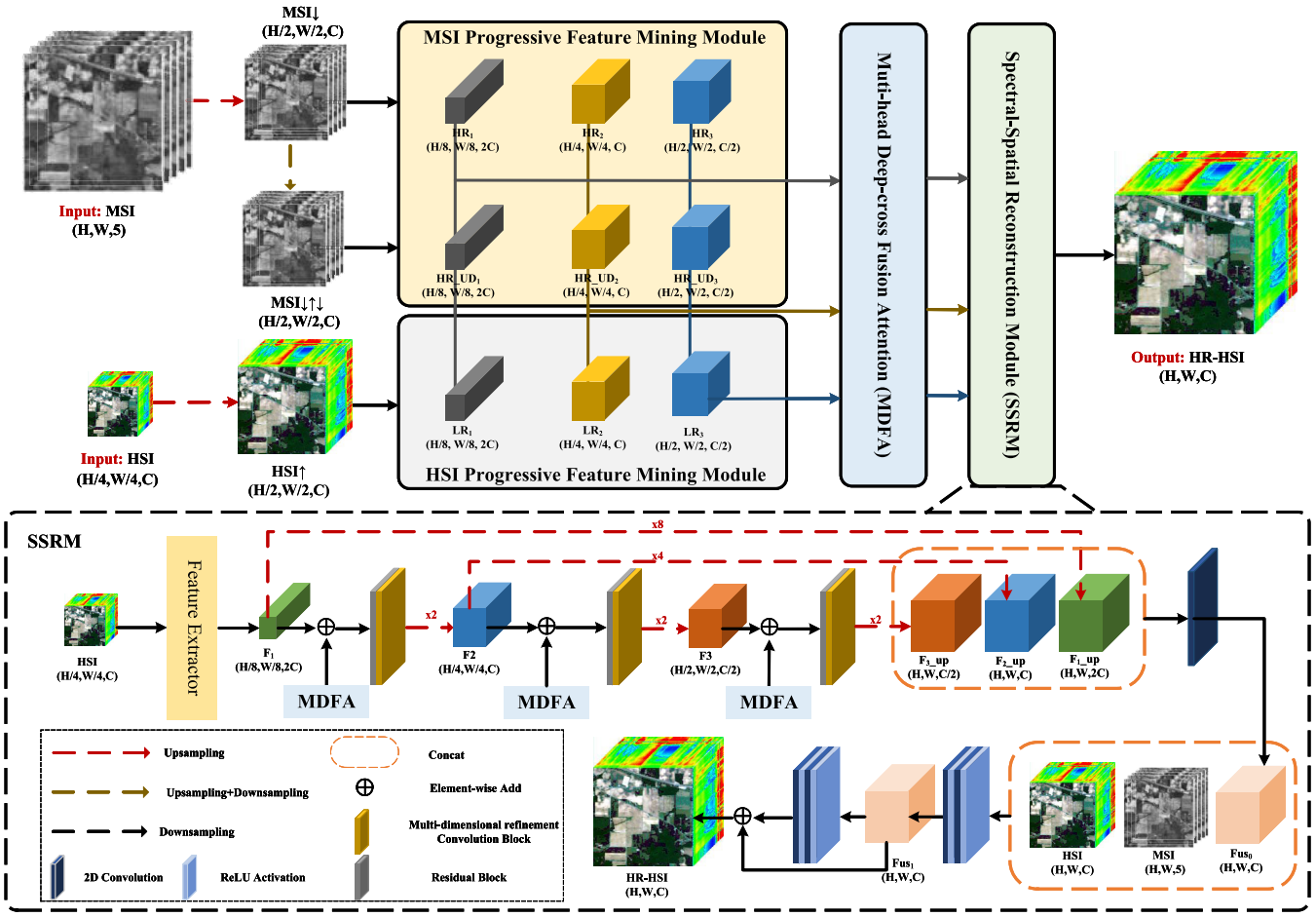


Fig. 1. Architecture of our proposed MDC-FusFormer algorithm.

and analysis. Section IV presents the conclusion and suggestions for future work.

II. Materials and Methods

Fig. 1 illustrates the general framework structure of the proposed MDC-FusFormer for HSI-MSI fusion. It contains three main parts: two independent PFMMs, an MDCFA module, and an SSFM. Among them, MSI ↓, MSI ↓↑↓, and HSI ↑ stand for MSI with twofold down-sampling, MSI with continuous twofold up-sampling and twofold down-sampling, and HSI with twofold up-sampling, respectively. We use bicubic interpolation to perform continuous up- and down-sampling operations on MSIs as well as up-sampling operations on HSIs. This is because it has been experimentally demonstrated that this helps HSI and MSI to produce smaller spatial and spectral distortions [53]. The successive up-sampling and down-sampling operations make the MSI ↓, MSI ↓↑↓, and HSI ↑ domains consistent, which is the key to the smooth and reliable operation of the subsequent attention mechanism.

In the proposed MDC-FusFormer, the estimated HR-HSI is defined as $R \in \mathbb{R}^{H\times W\times C}$, where H and W, respectively, represent the dimensions of the image's height and width, and C represents the number of spectral channels. In addition, we take LR-HSI and HR-MSI as inputs, denoted as $X \in \mathbb{R}^{h\times w\times C}$ and $Y \in \mathbb{R}^{H\times W\times c}$, respectively, where $h \ll H$, $w \leq W$, and $c \ll C$. Therefore, after the up-sampling and down-sampling process, MSI ↓, MSI ↓↑↓, and HSI ↑ are denoted as $Y_{\downarrow} \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, $Y'_{\downarrow} \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, and $X_{\uparrow} \in \mathbb{R}^{(H/2)\times(W/2)\times C}$, respectively.

A. Progressive Feature Mining Module

We first pass the processed $Y_{\downarrow}$ and $X_{\uparrow}$ through channel attention and spectral attention to improve the network's ability to perceive different features and better utilize the spectral information. After that, we take $X_{\uparrow}$, $Y_{\downarrow}$, and $Y'_{\downarrow}$ as three inputs and perform feature mining progressively.

We designed two independent PFMMs to extract the spectral and spatial features of HSI ↑, MSI ↓, and MSI ↓↑↓, as shown in Fig. 2.

Feature information at different scales is extracted by using convolutions with different kernel sizes. Specifically, the first layer successively uses a 7×7 convolution (stride $= 1$, padding $= 3$), a normalization layer, and an activation function. Then, a 3×3 convolution (stride $= 1$, padding $= 1$) and a normalization layer are used to extract features at the first scale. The recursive use of convolutional operations to progressively obtain different features

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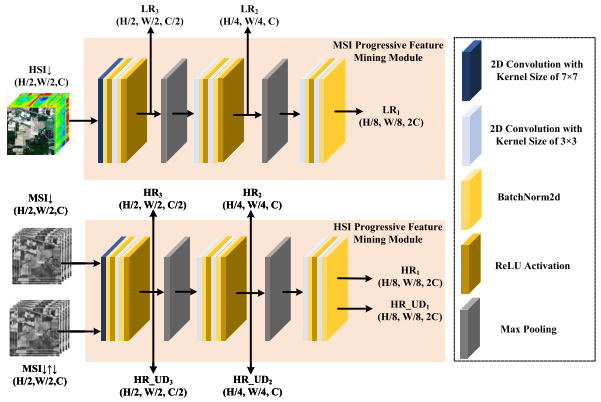


Fig. 2. Illustration of the PFMM.

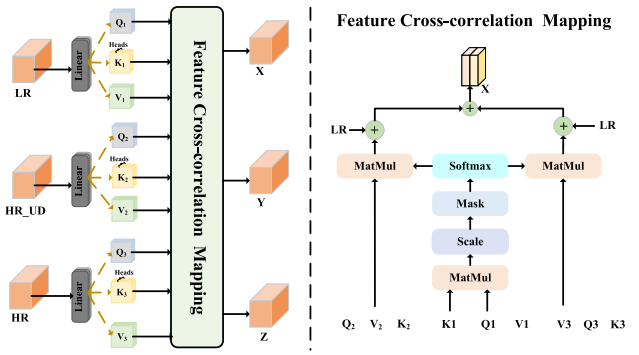


Fig. 3. Illustration of the MDCFA module.

of the input image from high resolution to low resolution is represented by the formula below using HSI ↑ as an example

$$\begin{aligned}
\mathrm{LR}_1 &= \mathrm{BN}\big(\mathrm{Conv}_{3\times3}\big(\mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv}_{7\times7}\big(X_{\uparrow}\big)\big)\big)\big)\big)\\
\mathrm{LR}_2 &= \mathrm{BN}(\mathrm{Conv}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(\mathrm{Maxpooling}\\
&\qquad (\mathrm{ReLU}(\mathrm{LR}_1)))))))\\
\mathrm{LR}_3 &= \mathrm{BN}(\mathrm{Conv}(\mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}(\mathrm{Maxpooling}\\
&\qquad (\mathrm{ReLU}(\mathrm{LR}_2))))))) \qquad (1)
\end{aligned}$$

where X represents the LR-HSI, $X_{\uparrow}$ represents the HSI image after up-sampling process, and LR is the extracted image features of HSI. LR_1, LR_2, and LR_3 are the features of HSI extracted at three different scales, respectively.

The rich local spectral and spatial information in the image is extracted in the process of gradual down-sampling. Then, the obtained three feature images of the same scale are input into the MDCFA to further simulate the long-distance dependence in global space and perform the interactive fusion of information across spatial and spectral domains.

B. MDCFA Module

To explore the long-range dependency in the global space as well as to realize the interactive fusion of spectral and spatial information, we designed a three-branch MDCFA module, as shown in Fig. 3, where $\mathrm{LR}_i(i = 1, 2, 3)$ stands for features extracted from LR-HSI at different scales, $\mathrm{HR_UD}_i(i = 1, 2, 3)$ denotes the features at different scales extracted from an MSI that has been successively up-sampled and down-sampled, and $\mathrm{HR}_i(i = 1, 2, 3)$ indicates the scale-features extracted from HR-MSI.

MDCFA adopts a three-branch structure for spatial and spectral information interaction. Each of the three branches contains different semantic information which can establish the dependency of the entire image on the three feature mappings and more fully realize the interaction of spatial and spectral information, as well as make full use of the spatial information of MSI. We first utilize the linear layer to generate the query (Q), key (K), and value (V) required for attention for all three inputs. Then, we compute the feature soft attention for each descriptor. Finally, we concatenate the output feature descriptors that have undergone feature soft attention and transform them back to the original feature space through the linear layer. The proposed fusion attention can greatly help the network to fully extract the dependencies of MSI and HSI across spatial and spectral feature spaces and fully utilize the spatially detailed features.

Feature Cross-Correlation Mapping: We do this by establishing the correlation between Q and K in the feature mapping. This involves computing the feature interrelationship matrices between the queries and keys for each of the three inputs separately and expressing it as a 3-D matrix $M \in \mathbb{R}^{N\times f_k\times f_q}$. Q_1, K_1, and V_1 are the queries, keys, and values generated by LR-HSI. Q_2, K_2, and V_2 are the queries, keys, and values generated by the MSI that has been sampled continuously up and down (MSI ↑↓). Q_3, K_3, and V_3 are the queries, keys, and values produced by HR-MSI. The characteristic relationships are calculated as follows (LR-HSI as an example):

$$M_1 = \mathrm{MatMul}\left(\frac{K_1}{\sqrt{d_k}}, Q_1^{\mathrm{T}}\right) \qquad (2)$$

where T represents the transpose operation, $d_k = (\mathrm{dim}/\mathrm{heads})$, dim is the feature dimension of the input, heads is the number of heads of multihead attention, and MatMul represents matrix multiplication. M_2 and M_3 represent the characteristic relationship matrices of MSI ↑↓ and HR-MSI, respectively. This can extract the respective feature interrelationships of HSI and MSI. Then, along the rows of each mutual matrix in M_1, we utilize a softmax layer to obtain the normalized mutual correlation matrix M_1', which we express as

$$M_1' = \mathrm{soft\,max}(M_1, \mathrm{dim} = 1). \qquad (3)$$

We use M_2' and M_3' to represent the correlation matrices of MSI ↑↓ and HR-MSI, respectively. They are calculated similar to that of M_1'. Then, we use the resulting matrix as a weight to calculate feature soft attention with the other two values (V) to perform feature fusion across spatial and spectral domains, as follows:

$$\begin{cases} I_1 = M_1' V_2 + \mathrm{LR} \\ I_2 = M_1' V_3 + \mathrm{LR}, \quad I = I_1 + I_2 \end{cases}$$
$$\begin{cases} J_1 = M_2' V_1 + \mathrm{HR_UD} \\ J_2 = M_2' V_3 + \mathrm{HR_UD}, \quad J = J_1 + J_2 \end{cases}$$
$$\begin{cases} L_1 = M_3' V_1 + \mathrm{HR} \\ L_2 = M_3' V_2 + \mathrm{HR}, \quad L = L_1 + L_2 \end{cases} \qquad (4)$$

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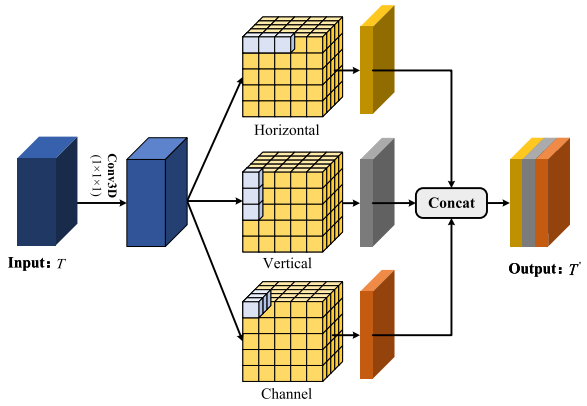


Fig. 4. Illustration of the MRCB.

where $I, J, and L$ are the final fusion features. Then, we reshape them through a linear layer and add them to get the final feature F, which is expressed as follows:

$$F = \text{Linear}(I) + \text{Linear}(J) + \text{Linear}(L). \tag{5}$$

This enables a more adequate interactive fusion of information between HSI and MSI across spectra and spatial spaces. In particular, the three-branch design provides rich and sufficient spatial information for the fusion task by introducing an up- and down-sampled MSI. The spatial features are more fully utilized, and the lack of spatial information is reduced to improve the effectiveness of fusion.

C. Spatial–Spectral Reconstruction Module

Different from fusing the spatial features of HSIs only at high-resolution scales, we extract the spatial features of MSIs at different scales. Specifically, we extract spatial details at three spatial scales: 1) the spatial scale of LR-HSI with twofold down-sampling; 2) the spatial scale of LR-HSI; and 3) the spatial scale of LR-HSI with twofold up-sampling of HSI.

We first connect the fused feature representation F obtained by MDCFA with the spectral feature G from the backbone network, as shown in Fig. 1. Next, we utilize 3×3 convolution and batch normalization (BN) layers to fuse the spatial–spectral information together and generate the residual components for the backbone. This residual component is then further used to generate HR-HSI. We define this process as follows:

$$\tilde{F} = \text{BatchNorm}(\text{Conv}(F + G)). \tag{6}$$

Then, we perform progressive up-sampling on the resulting fused image. The spatial–spectral detail information obtained through attention is combined with the input image in an up-sampling layer. This makes full use of the semantic features of the image, retains the spatial details of each band, and gradually reconstructs it into HR-HSI.

Multidimensional Refinement Convolution Block: Inspired by the one-dimensional transposed convolution in [54], we add a multidimensional convolution block to SSRM. Its structure is shown in Fig. 4. For the input feature image, the MRCB utilizes 3-D convolution to capture the spatial texture information from three dimensions: horizontal, vertical, and channel. Let $O \in \mathbb{R}^{H\times W\times C}$ be the input features. Firstly, O is input into the convolution, and then, by performing different dimensional transformations, it is input into the convolution of different scale convolution kernels to capture the features in the horizontal, vertical, and channel dimensions to refine the feature information. Eventually, the obtained features are spliced to get the final output. The MRCB can be represented as

$$\begin{aligned} &O_D[x, y, z] \\ &= (O * k)_D[x, y, z] \\ &= \sum_{m=-n}^{n} O[x + D_h m, y + D_w m, z + D_c m] \times k[n - m] \end{aligned} \tag{7}$$

where O is the input tensor and $O_D \in \mathbb{R}^{H\times W\times \tilde{C}}$ is the output tensor of the multidimensional refinement convolution. $[x, y, z]$ is each position of the input tensor O. $k \in \mathbb{R}^{2n+1}$ denotes a filter of size $2n + 1$, and $D = (D_h, D_w, D_c)$ denotes the dimension in which filter k is convolved. We have set $n = 1$, so that the size of k is 3. We use 3-D convolutions of size $3 \times 1 \times 1$, $1 \times 3 \times 1$, and $1 \times 1 \times 3$ to capture contextual information for each of the three dimensions of the feature.

D. Loss Function for MDC-FusFormer

We employed the straightforward and widely used root mean square error (RMSE) as the loss function, which yielded satisfactory convergence without the need for additional complexities in the network architecture. The RMSE is defined as

$$\text{Loss} = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{t=1}^{C}\left(\tilde{R}(i, j, t) - R(i, j, t)\right)^2}{2HWC} \tag{8}$$

where $\tilde{R} \in \mathbb{R}^{H\times W\times C}$ represents the reference LR-HSI.

E. Algorithm Summarization for MDC-FusFormer

The process of the proposed MDC-FusFormer network is shown in Algorithm 1.

III. Experiment and Analysis

In this section, we detail the dataset utilized for our experiments, the evaluation metrics, and the experimental setup. Subsequently, we present the results of the proposed MDC-FusFormer network on five hyperspectral datasets, which are then compared against those obtained from nine state-of-the-art methods. Additionally, we conduct ablation studies to validate the individual contributions and effectiveness of the various modules in our network. The source code for MDC-FusFormer is available at https://github.com/Nuist-HSI-Group/MDC-FusFormer.

A. Data Description

1) Pavia University: Pavia University was imaged by the airborne reflective optics spectrographic imaging system (ROSIS-3) during flights over Pavia University, Italy, in 2003. The resulting image has a spatial resolution of 1.3 m, a size

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Algorithm 1 MDC-FusFormer Network

Input:

HSI data $X \in \mathbb{R}^{h\times w\times C}$, MSI data $Y \in \mathbb{R}^{H\times W\times c}$ ($h \ll H, w \leq W, c \ll C$), ground-truth $\tilde{R} \in \mathbb{R}^{H\times W\times C}$.

Output:

The estimated HSI data $R \in \mathbb{R}^{H\times W\times C}$.

1: Perform up- and down-sampling operations on HSI and MSI to obtain three inputs: $HSI\uparrow$, $MSI\downarrow$, and $MSI\downarrow\uparrow\downarrow$.
2: The three images are fed into two independent PFMMs to extract feature images at three different scales for HSI and MSI, respectively through (1). These three scales are 1): the spatial scale of twofold down-sampled HSI; 2): the spatial scale of HSI, and 3): the spatial scale of HSI with twofold up-sampling.
3: Features at the same scale are input into MDCFA two by two to compute feature correlations. Interactive fusion of features across spatial and spectral domains through (2)-(4), which is performed by learning non-local dependencies to obtain the fusion feature image F at each scale through (5).
4: The feature representation F is summed with the feature image G of the feature-extracted HSI in the backbone network by (6).
5: The multidimensional refinement convolution is used in SSRM to spatially refine the features through (7) in three dimensions, *i.e.*, horizontal, vertical, and channel to obtain a feature image after texture feature refinement.
6: Finally, combining cascaded up-sampling as well as residual concatenation, the feature images at each scale are gradually fused to reconstruct HR-HSI images.

of 610 × 340 pixels, and 115 spectral bands. Twelve of these bands were eliminated due to the influence of noise; we selected the remaining 103 spectral bands for the experiment.

2) Washington DC: The Washington DC dataset is an aerial hyperspectral image of the sky over the Washington Mall taken by the HYDICE sensor in 1995. The dataset contains 191 bands in the visible and near-infrared wavelength ranges. The data size is 1208 × 307.

3) Salinas: The Salinas dataset is a set of images taken by the airborne visual infrared imaging spectrometer (AVIRIS) sensor in the Salinas Valley, California. The spatial resolution of the data is 3.7 m and the size is 512 × 217. The raw dataset has 224 bands; after removing the bands with heavy water vapor absorption, 204 bands remain.

4) Pavia Center: The Pavia Center dataset was acquired by the ROSIS sensor during flights over Pavia, northern Italy, in the spectral range 430–860 nm, with a total of 115 bands and a spatial resolution of 1.3 m. For practical use, 102 bands with a size of 1096 × 715 were selected for training.

5) Houston: The Houston dataset was acquired by the ITRES CASI-1500 sensor in Houston, Texas, USA, and its surrounding rural areas. The spatial resolution is 2.5 m and the data size is 349 × 1905. The band range is 364–1046 nm, with a total of 144 bands.

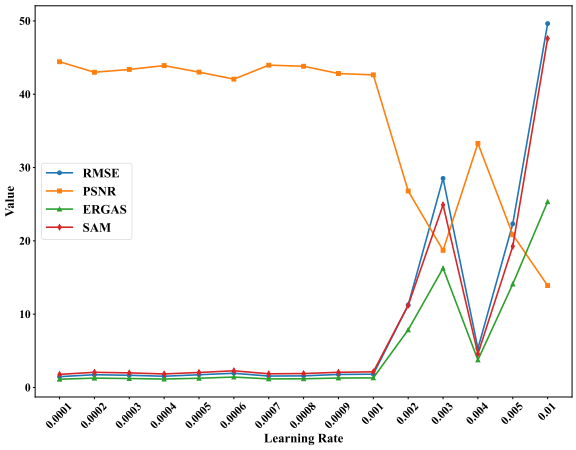


Fig. 5. Comparison of model effects under different learning rates. The x-axis represents different learning rates; the y-axis corresponds to the values of RMSE, PSNR, ERGAS, and SAM at each learning rate.

B. Experimental Setting

1) Data Collection: For these five datasets, we simulate the LR-HSI and HR-MSI from the HR-HSI. We regard the LR-HSI and HR-MSI as the subsampling of the HR-HSI, from which we simulate the LR-HSI and HR-MSI. Firstly, we perform a Gaussian filtering of 5 × 5 on the HR-HSI in the dataset to get the blurred HSI. Then, we down-sample it four times to generate the low-resolution HSI, that is the LR-HSI. Finally, we take five images from HR-HSI at regular intervals as HR-MSI.

We take the 128 × 128 region in the center of the image as our test set and the rest as the training region. Before training, we fill the test region with zeros to ensure that our training covers the entire region. The test set and the training set are non-overlapping. Then, we randomly take a region of size 128 × 128 from the training region for training.

2) Evaluation Criteria: To quantitatively analyze the effectiveness of the proposed method in comparison with other methods, we use five evaluation metrics: peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM), RMSE, erreur relative globale adimensionnelle de synthèse (ERGAS), and structural similarity (SSIM). PSNR and SSIM are positively correlated indicators, where higher values indicate better quality of the fused image. In contrast, SAM, RMSE, and ERGAS are negatively correlated indicators, with smaller values indicating better fusion quality.

3) Environment Configuration: The experiments are all conducted in the PyTorch environment using the processor Intel[1] Xeon[1] Silver 4314, 2.40 GHz, and an NVIDIA GeForce RTX 4090 GPU server. Adam is used as the optimizer, and the initial learning rate is set to 1e−4. The experiments are verified as shown in Fig. 5, which indicates that, when the initial learning rate is set to 1e−4, the fusion effect is optimal.

C. Ablation Study

To demonstrate the effectiveness of the various modules within the MDC-FusFormer, we conducted ablation studies

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TABLE I
EXPERIMENTAL RESULTS ON THE HOUSTON DATASET FOR DIFFERENT NUMBERS OF INPUTS TO THE ATTENTION MODULE

Cases	Houston				
	Component	Indicators			
		RMSE	PSNR	ERGAS	SAM
1	MDC-FusFormer with no attention	1.0329	46.4575	1.5875	1.7448
2	MDC-FusFormer with one input	1.0114	47.6401	1.4377	1.7319
2	MDC-FusFormer with two inputs	0.9950	47.7823	1.6668	1.7631
3	MDC-FusFormer	**0.8056**	**49.6162**	**1.1814**	**1.4818**

TABLE II
EXPERIMENTAL RESULTS OF MRCB MODULES OF DIFFERENT DIMENSIONS ON PAVIA UNIVERSITY DATASET

Cases	Pavia University				
	Component	indicators			
		RMSE	PSNR	ERGAS	SAM
1	without MRCB	1.7681	42.8657	1.2920	2.0677
2	MRCB(two-dimensional)	1.7675	42.8687	1.2846	2.0921
3	MRCB	**1.4606**	**44.5251**	**1.1178**	**1.7679**

on the Houston dataset, examining the model's performance with different configurations of these modules.

1) Deep Cross-Fusion Attention Ablation Studies: Deep cross-fusion attention in the MDC-FusFormer model plays an important role in image reconstruction. A three-branch structure was designed to capture the global feature correlation of the three features by computing the dependency between Q, K, and V for the other two features two by two to perform the interaction of spatial–spectral features for deep fusion. The interactively fused features were finally obtained. We conducted experiments on the Houston dataset with three different cases: the case without the attention module, the one-input attention case, the normal two-input attention case, and the three-input attention case. We evaluated the impact on the model's performance using four metrics: RMSE, PSNR, ERGAS, and SAM. The results are shown in Table I.

As observed in Table I, the three-branch attention mechanism is more adept at interacting with spatial and spectral information, facilitating the fusion of HSI and MSI and resulting in reduced spatial and spectral distortions. Compared with the two-input attention, it reduces the RMSE, ERGAS, and SAM metrics by 23.5%, 29.1%, and 16.0%, respectively, and improves the PSNR by 3.8%. This fully proves that the deep fusion attention of the three branches can well perform the interactive fusion of spatial and spectral information while preserving good spatial details.

2) Multidimensional Refinement Convolution Block Ablation Studies: The edge information of an image contains high-frequency features, which are important for reconstructing high-resolution images. Therefore, we designed multidimensional refinement convolutional blocks to perform the spatial refinement of edge features in three dimensions: horizontal, vertical, and channel. The spatial refinement of the fused features was performed by setting different sizes of convolution kernels in the three dimensions. For this, we conducted the corresponding ablation experiments on the Pavia University dataset, as detailed in Table II.

TABLE III
EXPERIMENTAL RESULTS ON CONVOLUTIONAL KERNEL SIZES FOR THE MRCB MODULE ON THE PAVIA UNIVERSITY DATASET

Cases	Pavia University				
	Number of convolution kernels	Indicators			
		RMSE	PSNR	ERGAS	SAM
1	3	**1.4606**	**44.5251**	**1.1178**	**1.7679**
2	5	1.5142	44.2119	1.1567	1.8200
3	7	1.5072	44.2520	1.1448	1.8208

From Table II, it is clear that the fusion quality of the image is improved by adding multidimensional refinement convolution. ERGAS is improved by 12%. Compared to the 2-D multidimensional refinement convolution, the 3-D multidimensional refinement convolution can better refine the feature details and reconstruct higher-quality images. Meanwhile, we also experimented with the choice of convolution kernel size on the Pavia University dataset, and the results are shown in Table III. It can be seen that the PSNR value reaches a maximum at a convolution kernel size of 3 and a minimum in RMSE, ERGAS, and SAM metrics. Therefore, we choose convolution kernel size 3 for 3-D convolution to refine the features and get the higher-quality reconstructed image.

Combined with the above ablation experiments, the role of each part of the MDC-FusFormer was fully demonstrated, and the effectiveness of the model in HSI-MSI fusion as well as image reconstruction was verified.

D. Comparison With State-of-the-Art Methods

To quantitatively evaluate the fusion effect of the MDC-FusFormer model, we used nine state-of-the-art methods for comparison: two traditional methods, that is, CNMF [4] and GSA [55], and seven deep learning methods, that is, MSDCNN [36], TFNet [56], ResTFNet [56], SSFCNN [57], ConSSFCNN [57], SSR-NET [39], and MCT-NET [58].

1) Pavia University Results: The experimental results are reported in Table IV, where the best results are shown in bold. To show the fusion effect more intuitively, we present the pseudo-color images of the fused HR-HSI and the difference images between the reference image and the estimated image obtained by all the methods in Fig. 6. From Table IV, we can see that the proposed methods are optimal in RMSE, PSNR, ERGAS, and SAM. The deep learning methods are all better than the traditional learning methods. This is because deep learning methods can fully extract image feature information using CNNs. Both the MCT-NET and MDC-FusFormer methods outperform other methods. This is because both methods not only focus on local features but also use the transformer for extracting global dependencies. However, the MDC-FusFormer model is designed with a three-branch deep

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TABLE IV

FUSION PERFORMANCE OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY, WASHINGTON DC, AND SALINAS DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Pavia University					Washington DC					Salina				
	RMSE	PSNR	ERGAS	SAM	SSIM	RMSE	PSNR	ERGAS	SAM	SSIM	RMSE	PSNR	ERGAS	SAM	SSIM
CNMF [4]	7.5816	30.5356	3.8939	2.2469	0.9176	6.7624	38.6050	3.1321	1.7065	0.5960	3.2888	36.9072	8.0077	7.9419	0.6685
GSA [55]	4.4483	35.1671	2.2940	4.5822	0.9462	3.7713	36.6009	3.2343	1.8035	0.5411	14.2258	26.0058	10.7557	4.0942	0.6685
MSDCNN [36]	2.2960	40.5965	1.6400	2.5872	0.9685	2.6779	36.9109	0.4686	0.8914	0.5425	2.9781	37.9935	3.7010	8.5247	0.7484
TFNet [56]	2.2366	40.8240	1.6033	2.4720	0.9689	1.9704	39.5756	0.346	0.6756	0.6116	1.9187	41.8121	2.5821	1.4839	0.8277
ResTFNet [56]	2.0105	41.7495	1.4783	2.3185	0.9723	1.7291	40.7100	0.3017	0.5816	0.6523	2.2631	40.3784	3.4017	1.8130	0.7816
SSFCNN [57]	1.9410	42.0550	1.3957	2.2121	0.9651	16.4613	21.1373	3.1689	6.8978	0.6460	7.2907	30.2170	10.2749	7.5414	0.7272
ConSSFCNN [57]	2.3572	40.3676	1.6150	2.4065	0.9737	14.0913	22.4876	2.6805	5.8734	0.5462	7.3050	30.2000	15.4438	7.3377	0.4564
SSR-NET [39]	1.6300	43.5722	1.2065	1.9517	0.9779	2.0591	39.1928	0.3603	0.7201	0.6149	2.5673	39.2828	7.2998	1.8444	0.7367
MCT-NET [58]	1.5580	43.9644	1.1685	1.8684	0.9787	0.893	46.4498	0.1553	0.2983	0.8091	2.2920	40.2680	7.0344	1.8393	0.7783
MDC-FusFormer	**1.4606**	**44.5251**	**1.1178**	**1.7679**	**0.9821**	**0.7778**	**47.6495**	**0.1352**	**0.2626**	**0.8595**	**1.7626**	**42.5491**	**5.3816**	**1,3676**	**0.8437**

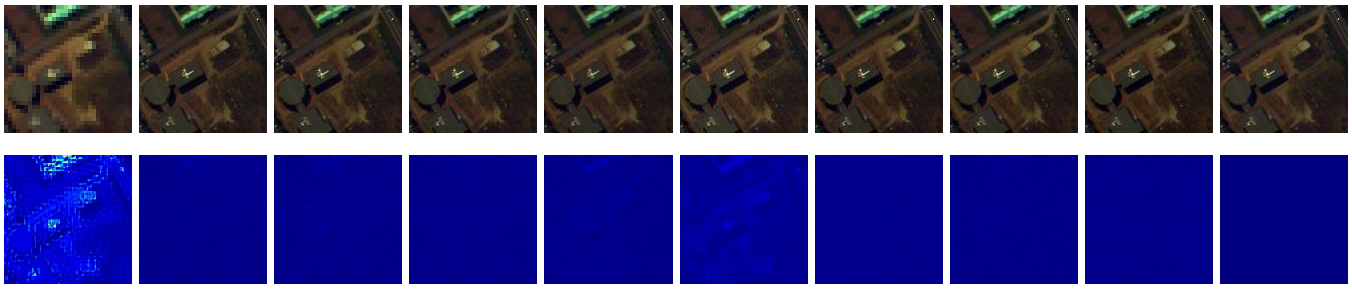


Fig. 6. Fusion results of different methods on the Pavia University dataset. The first line is the R-G-B image of the estimated HR-HSI (67-29-1 band), and the second line is the pseudo-color difference image of the estimated R-G-B image and the reference image. (a) LR-HSI. (b) MSDCNN. (c) TFNet. (d) ResTFNet. (e) SSFCNN. (f) ConSSFCNN. (g) SSR-NET. (h) MCT-NET. (i) MDC-FusFormer. (j) GT.

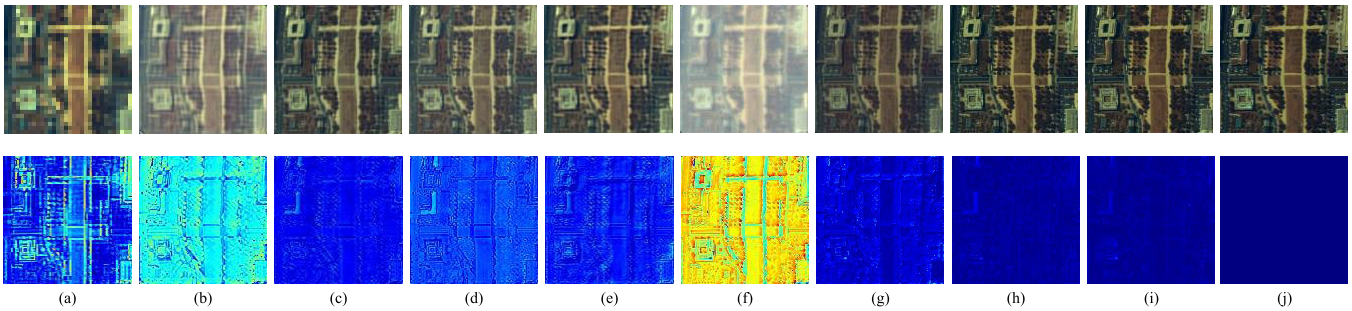


Fig. 7. Fusion results of different methods on the Washington DC dataset. The first line is the R-G-B image of the estimated HR-HSI (55-35-11 band), and the second line is the pseudo-color difference image of the estimated R-G-B image and the reference image. (a) LR-HSI. (b) MSDCNN. (c) TFNet. (d) ResTFNet. (e) SSFCNN. (f) ConSSFCNN. (g) SSR-NET. (h) MCT-NET. (i) MDC-FusFormer. (j) GT.

cross-attention, which can capture the global complementary information more fully for the interaction of spatial–spectral information. As a result, it reduces the RMSE, ERGAS, and SAM metrics by 6.3%, 4.3%, and 5.4%, respectively, compared to MCT-NET, and improves the PSNR by 1.3%.

2) Washington DC Results: The experimental results are reported in Table IV. Fig. 7 demonstrates the fused HR-HSI pseudo-color image and the difference image between it and the reference image. It can be seen that the proposed method achieves excellent results on this dataset. The PSNR value is improved by 2.6% compared to MCT-NET and by 21.58% compared to SSR-NET. This proves that MDC-FusFormer has a good ability to preserve spatial details. Although the Washington DC dataset has a very large number of bands, MDC-FusFormer also achieves good results in the SAM and ERGAS metrics. This confirms its ability to retain spectral and spatial information well with small spectral and spatial distortions. Additionally, we show the spectral curves of the fused images of the four methods with ground truth (GT) in Fig. 8(a). It can be seen from the zoomed-in area in the figure that our method can reconstruct the high-resolution image well, bringing it very close to GT.

3) Salinas Results: Table IV reports the comparison results between the proposed method and the comparison methods tested on the Salinas dataset. The corresponding fused R-G-B images are shown in Fig. 9. It is evident that the MDC-FusFormer outperforms all other methods in terms of RMSE, PSNR, ERGAS, and SAM. For the ERGAS metric, our method achieves the best results when compared to other state-of-the-art techniques. This demonstrates

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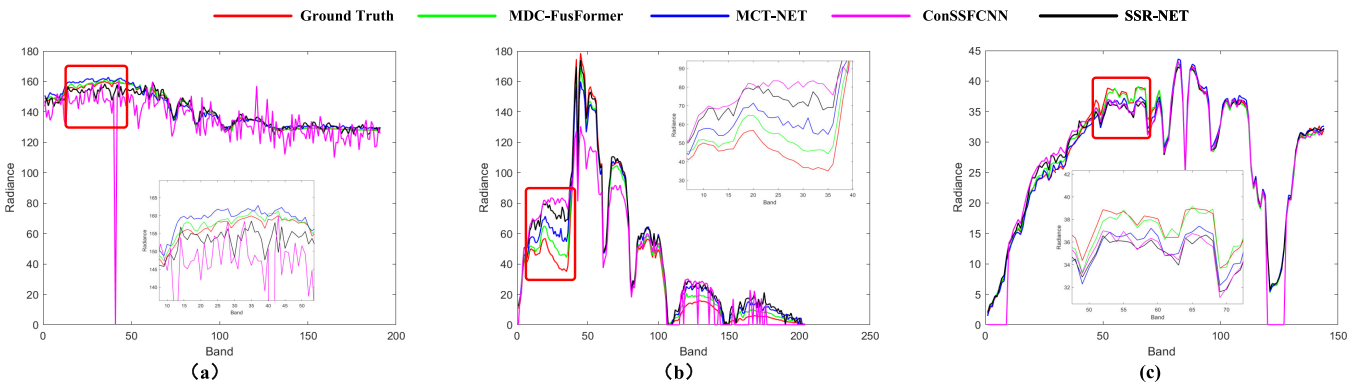


Fig. 8. Comparison of spectral profiles at a specific location on the Washington DC, Salinas, and Houston datasets. (a) Washington DC (127, 127). (b) Salinas (70, 70). (c) Houston (100, 100).

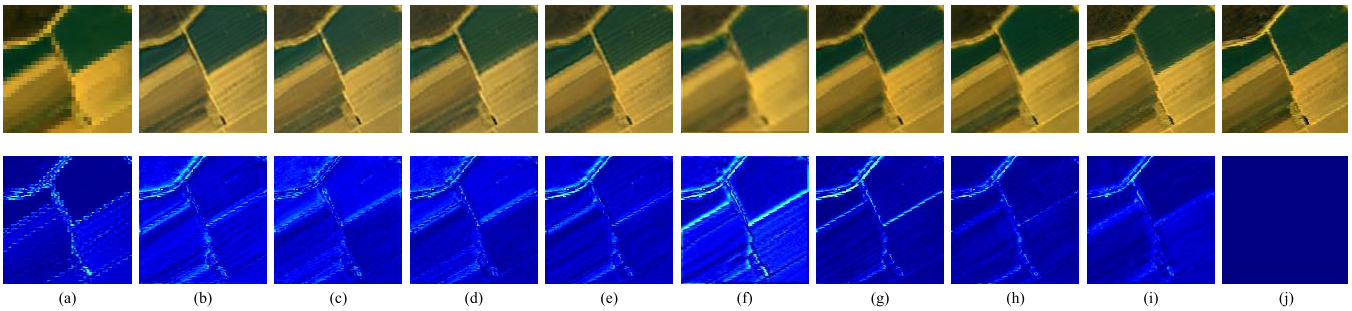


Fig. 9. Fusion results of different methods on the Salinas dataset. The first line is the R-G-B image of the estimated HR-HSI (29-15-4 band), and the second line is the pseudo-color difference image of the estimated R-G-B image and the reference image. (a) LR-HSI. (b) MSDCNN. (c) TFNet. (d) ResTFNet. (e) SSFCNN. (f) ConSSFCNN. (g) SSR-NET. (h) MCT-NET. (i) MDC-FusFormer. (j) GT.

TABLE V

FUSION PERFORMANCE OF DIFFERENT METHODS ON THE PAVIA CENTER DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Pavia Center				
	RMSE	PSNR	ERGAS	SAM	SSIM
CNMF [4]	13.978	25.2221	11.436	4,3635	0.8958
GSA [55]	5.0793	34.4197	5.6044	5.8472	0.9665
MSDCNN [36]	4.1771	35.7134	4.7038	5.2175	0.9738
TFNet [56]	4.2554	35.5520	4.8124	4.8567	0.9752
ResTFNet [56]	3.7568	36.6343	4.2977	4.3930	0.9794
SSFCNN [57]	3.7408	36.6714	4.3153	4.2893	0.9812
ConSSFCNN [57]	4.5431	34.9837	4.9532	4.7606	0.9731
SSR-NET [39]	3.2632	37.8579	3.7296	3.7891	0.9818
MCT-NET [58]	2.7284	39.4127	3.0898	3.5273	0.9844
MDC-FusFormer	**2.5017**	**40.1660**	**2.8742**	**3.3535**	**0.9874**

TABLE VI

FUSION PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Houston				
	RMSE	PSNR	ERGAS	SAM	SSIM
CNMF [4]	1.1720	47.8682	1.6965	1.2262	0.9826
GSA [55]	1.4246	45.0846	1.9642	2.0723	0.9789
MSDCNN [36]	1.9684	41.8563	2.3205	3.4821	0,9682
TFNet [56]	1.4676	44.4061	1.7938	2.5231	0,9801
ResTFNet [56]	1.1361	46.6301	1.5268	1.9726	0,9861
SSFCNN [57]	1.3990	44.8223	4.0564	2.6784	0,9661
ConSSFCNN [57]	3.5016	36.8531	9.4206	6.5112	0,8522
SSR-NET [39]	1.3702	45.0026	1.8997	2.5191	0,9766
MCT-NET [58]	1.3488	45.1395	2.0394	2.4682	0,9763
MDC-FusFormer	**0.8056**	**49.6162**	**1.1814**	**1.4818**	**0.9912**

that the proposed approach emphasizes both local information and global features. Moreover, it can be seen from the spectral curve comparison in Fig. 8(b) that the spectral curve of MDC-FusFormer is closer to GT and the reconstruction quality is higher compared to the other methods.

4) Pavia Center Results: The experimental results of the proposed MDC-FusFormer method on this dataset are reported in Table V. To be able to visualize the quality of the evaluated fusion, we show the fused HR-HSI pseudo-color image and the difference image between it and the reference image in Fig. 10. The evaluation results clearly show that the proposed method performs the best. MDC-FusFormer achieves optimal values for the RMSE, PSNR, ERGAS, and SAM metrics. It has obvious advantages in terms of spatial reconstruction quality, elemental reconstruction quality, and spectral reconstruction quality.

5) Houston Results: The results of the method proposed in this article on the Houston dataset are reported in Table VI. The corresponding R-G-B images are shown in Fig. 11. From

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TABLE VII
COMPARISON RESULTS OF FLOPS AND PARAMETERS OF THE PROPOSED MDC-FUSFORMER MODEL WITH OTHER METHODS. THE BEST RESULTS ARE SHOWN IN BOLD

Method	Pavia University						
	Params (M)	Flops (G)	Test time (ms)	RMSE	PSNR	ERGAS	SAM
MSDCNN [36]	1.25	20.52	6.74	2.2960	40.5965	1.6400	2.5872
TFNet [56]	2.45	9.11	5.94	2.2366	40.8240	1.6033	2.4720
ResTFNet [56]	2.33	8.47	5.76	2.0105	41.7495	1.4783	2.3185
SSFCNN [57]	0.46	7.55	5.09	1.9410	42.0550	1.3957	2.2121
ConSSFCNN [57]	0.48	7.84	4.98	2.1667	41.0999	1.5057	2.2972
SSR-NET [39]	**0.29**	**4.69**	**4.51**	1.6300	43.5722	1.2065	1.9517
MCT-NET [58]	4.07	13.28	28.89	1.5580	43.9644	1.1685	1.8684
MDC-FusFormer	17.79	24.29	41.94	**1.4768**	**44.4296**	**1.1294**	**1.7862**

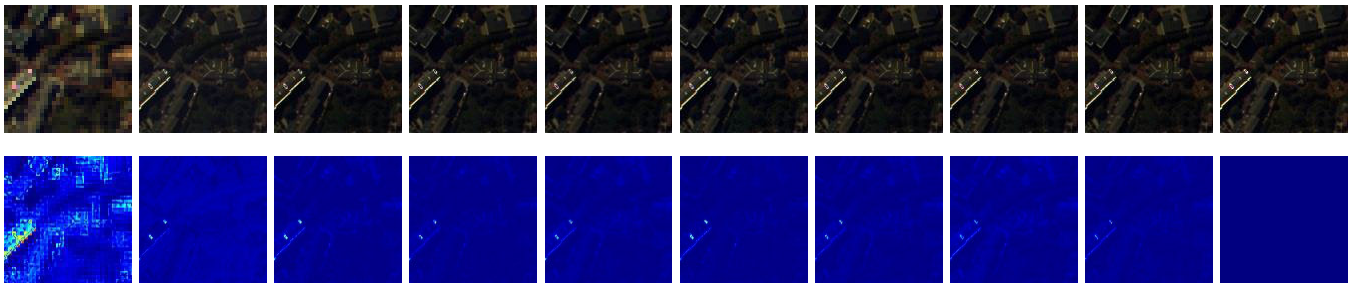


Fig. 10. Fusion results of different methods on the Pavia Center dataset. The first line is the R-G-B image of the estimated HR-HSI (67-29-1 band), and the second line is the pseudo-color difference image of the estimated R-G-B image and the reference image. (a) LR-HSI. (b) MSDCNN. (c) TFNet. (d) ResTFNet. (e) SSFCNN. (f) ConSSFCNN. (g) SSR-NET. (h) MCT-NET. (i) MDC-FusFormer. (j) GT.

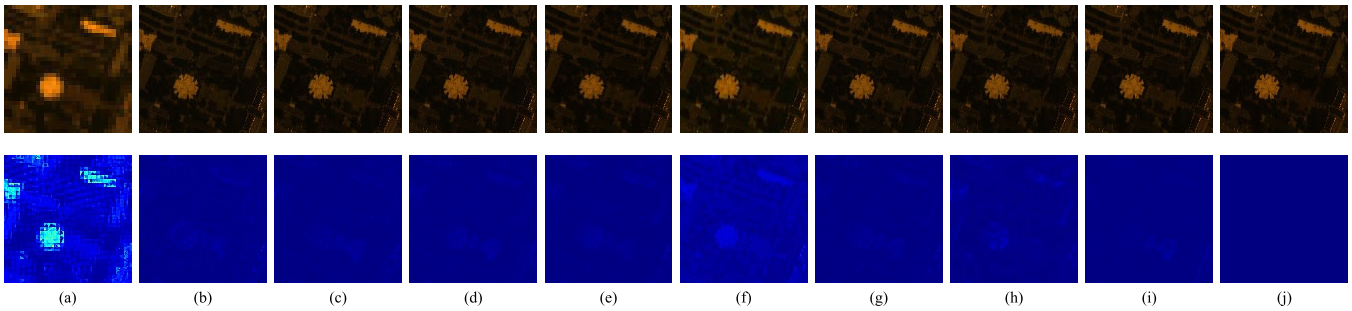


Fig. 11. Fusion results of different methods on the Houston dataset. The first line is the R-G-B image of the estimated HR-HSI (29-15-4 band), and the second line is the pseudo-color difference image of the estimated R-G-B image and the reference image. (a) LR-HSI. (b) MSDCNN. (c) TFNet. (d) ResTFNet. (e) SSFCNN. (f) ConSSFCNN. (g) SSR-NET. (h) MCT-NET. (i) MDC-FusFormer. (j) GT.

the table, it can be seen that our method performs best on all four evaluation metrics. A great improvement is achieved compared to MCT-Net. This demonstrates the ability of our method to extract and preserve spatial details and improve the fusion quality in general. Fig. 8(c) shows a comparison plot of the spectral curves of the four fusion methods on the Houston dataset. The near overlap between the spectral curves of MDC-FusFormer and GT can be clearly seen in the zoomed-in region, confirming the superiority of MDC-FusFormer over the other methods.

E. Analysis of Model Complexity

To assess the time efficiency, parameter count, and model complexity of the tested methods, we present in Table VII the test time, number of parameters, and computational complexity of various models on the Pavia University dataset. The values of the PSNR, SAM, and ERGAS metrics for different methods are also listed. Floating point operations per second (FLOPs) is a standard measure of a computer system or an algorithm's performance, indicating the number of floating point operations executed per second. In deep learning, FLOPs is frequently used to gauge the computational complexity of a neural network model, that is, the number of floating point operations required for forward propagation. A higher FLOPs value suggests greater computational resource demands, which may correlate with improved performance and accuracy. Since both HSI and MSI have a large amount of data and we have used three-branch deep cross-fusion attention for computing the correlation between features, the computational effort has increased considerably and the number of parameters has also risen. However, the quality

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of the reconstructed image is also improved. In comparison to SSR-NET, which has the simplest model and the fewest parameters, the MDC-FusFormer model is more intricate and has a larger parameter set. However, it also yields a considerably superior image reconstruction quality. The table reveals a 9.4% decrease in RMSE, a 2.0% improvement in PSNR, a 6.4% decrease in ERGAS, and an 8.5% decrease in SAM when using the MDC-FusFormer model.

IV. Conclusion

In this article, a transformer-based multiscale deep cross-fertilization network (MDC-FusFormer) is proposed for hyperspectral and multispectral image fusion. The MDC-FusFormer mainly consists of two PFMMs, a deep cross-attention fusion module, and an SSFM. A processed hyperspectral image is added as input to provide sufficient spatial information to reduce spatial distortion. By computing the correlation between different features in a two-by-two manner, non-local complementary features are captured. Deep interaction between HSI and MSI across spatial and spectral domains is achieved. Meanwhile, multidimensional refinement convolution is used to refine the features of the fused image and improve the performance. We conducted an extensive array of experiments across five datasets; our model demonstrated superior performance when compared to other state-of-the-art methods.

However, our current model primarily concentrates on the fusion of well-aligned images. In real-world scenarios, there might be significant misalignments between HSI and MSI. Additionally, our model incorporates a substantial number of parameters. In our future research, we intend to address these challenges by developing lightweight fusion models that not only reduce the parameter count but also enhance the fusion performance.

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